MILITARY SPECIFICATION

BAROSWITCH M13

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of switch assembly entitled Baroswitch M13 (BSA) which is used in the Sergeant warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification for.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing & Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE

8822263	- Baroswitch M13.
8853654	- Drift and Leakage Test Procedures.
EL-8871314	- Equipment List.

FSC: 1135

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From Terminal	To Terminal	Resistance (ohms)	Test Code Nos.
5J1-G	5J1-C or D	0.4 max.	030001
5J1-G	5J1-E or F	0.4 max.	030002
5J1-M	5J1-H or J	0.4 max.	030003
5J1-M	5J1-K or L	0.4 max.	030004

3.6.2 Miscellaneous circuits. The resistance of the miscellaneous circuits shall not exceed 0.1 ohms between the connector pins listed when tested as specified in 4.4.1.2.

From pin	To pin	From pin	To pin
A	B	H	J
C	D	K	L
E	F	M	G

3.7 Insulation resistance. The insulation resistance between electrically isolated circuits and between ungrounded circuits and the case, as listed, shall be greater than 25 megohms with a test voltage of 470 ± 30 volts direct current (VDC) when tested as specified in 4.4.2.

From pin	To pin	From pin	To pin
A	Ground	C	E
C	Ground	C	H
E	Ground	C	K
H	Ground	C	G
K	Ground	E	H
G	Ground	E	K
A	C	E	G
A	E	H	K
A	H	H	G
A	K	K	G
A	G		

3.8 Pneumatic system sealing. With the pneumatic system subjected to a pressure of 500 millimeter (mm) of mercury (Hg) less than ambient, the pressure change rate within the pneumatic system of the BSA shall not exceed 3 mm of Hg per minute (Min.) at ambient temperature for test system volume of 10 cubic inches when tested as specified in 4.4.3. At -65 ± 5 degrees Fahrenheit (°F.) temperature, the pressure change rate shall not exceed 6 mm Hg per Min. when tested as specified in 4.4.4.

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Table I

Operational tolerance and repeatability tests and codes

Test	Test Code Nos.			
Temperature: 80 ± 20°F.				
Test Run / Pressure (MM of Hg)	Elements - Pressure (MM Hg) 1	2	3	4
1 — 178.8 ± 10.45	060001	060002	060003	060004
2 — 178.8 ± 10.45	060005	060006	060007	060008
3 — 178.8 ± 10.45	060009	060010	060011	060012
Max. Diff ± 2.0				
Temperature: +160°F.				
Test Run / Pressure (MM of Hg)	Elements - Pressure (MM Hg) 1	2	3	4
1 — 178.8 ± 10.45	060013	060014	060015	060016
2 — 178.8 ± 10.45	060017	060018	060019	060020
3 — 178.8 ± 10.45	060021	060022	060023	060024
Max. Diff ± 2.0				
Temperature: -65°F.				
Test Run / Pressure (MM of Hg)	Elements - Pressure (MM Hg) 1	2	3	4
1 — 178.8 ± 10.45	060025	060026	060027	060028
2 — 178.8 ± 10.45	060029	060030	060031	060032
3 — 178.8 ± 10.45	060033	060034	060035	060036
Max. Diff ± 2.0				

3.10 Repeatability. At ambient, -65 ± 5°F. and 160 ± 5°F temperatures, the maximum difference between the operating points for any one element shall not exceed 2.0 MM Hg for three or more simulated climbs in any 24 hour period when tested as specified in 4.4.8, 4.4.9 and 4.4.10. If any element fails the repeatability test, record this information along with the code as listed in Table I with the following exception: Change the first three digits to 070. The last three digits remain the same.

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Frequency range (Hz)	Cycle time	Constant acceleration or double amplitude
8-55-8	2 min. ± 10 sec.	0.060 ± 0.006 inch
55-500-55	13 ± 1 min.	10 ± 1 g

3.11.7 Operational vibration. The BSA shall withstand a programmed sequence of band-limited white Gaussian noise and combined noise and sine wave vibration without evidence of damage or affecting functioning when tested as specified in 4.4.11.7. There shall be no evidence of contact chatter during the applied vibrations. The total time of test shall be 420 seconds; two tests of 70 seconds in each of three mutually perpendicular axes, as shown in Figure 3. The complex vibration test consists of the following five sections programmed continuously:

a. White Gaussian noise at a level of 12 g root mean square (RMS) band-limited between 15 and 1500 Hz for one second.
b. A 2.2 g RMS swept sine wave superimposed on 4.1 g RMS of white Gaussian noise band-limited between 15 and 1500 Hz for 15 seconds. The sine wave shall be swept from 800 to 1300 Hz at a rate increasing directly with frequency, one such sweep taking the full 15 seconds.
c. A 4.4 g RMS swept sine wave superimposed on 8.2 g RMS of white Gaussian noise band-limited between 15 and 1500 Hz for 15 seconds. The sine wave shall be swept from 1300 to 800 Hz at a rate decreasing directly with frequency, one such sweep taking the full 15 seconds.
d. A 2.2 g RMS sine wave superimposed on 4.1 g RMS of white Gaussian noise band-limited between 15 and 1500 Hz for 38 seconds. The sine wave shall be swept from 15 to 1500 Hz and back to 15 Hz at a rate increasing directly with frequency. One such cycle shall take the full 38 seconds.
e. White Gaussian noise at a level of 12 g RMS band-limited between 15 and 1500 Hz for one second.

Following the test the BSA shall meet the requirements of 3.6, 3.7, 3.8 at ambient temperature, and 3.9, at ambient.

3.11.8 Humidity. The BSA shall withstand exposure to high humidity cycles without deleterious effect when tested as specified in 4.4.11.8. After exposure to humidity cycles, the BSA shall comply with the requirements of 3.6, 3.7, 3.8 at ambient temperature, and 3.9, at ambient.

3.12 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

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4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table II. Tests shall be conducted in order specified in Table II.

Table II

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Baroswitch elements (see 3.6.1)		X
3. Miscellaneous circuits (see 3.6.2)		X
4. Insulation resistance (see 3.7)		X
5. Pneumatic system sealing (see 3.8)		X
6. Operational tolerance (see 3.9)		X
7. Repeatability (see 3.10)		X
8. Overpressure (see 3.11.1)	X	
9. Underpressure (see 3.11.2)	X	
10. Acceleration (see 3.11.3)	X	
11. Mechanical shock (see 3.11.4)	X	
12. Pressure cycling (see 3.11.5)	X	
13. Transportation vibration (see 3.11.6)	X	
14. Operational vibration (see 3.11.7)	X	
15. Humidity (see 3.11.8)	X	
16. Drift and leakage (see 3.13)	X	

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

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Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Major:
Isolator moldings
101. Excessive flash not removed.....Visual
102. Presence of dirt or chips either loose or embedded in isolator.....Visual

Minor: None defined.

4.3.2.1.2 Baroswitch, inspection of final assembly (see dwg 8822263).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Wires not connected to plug or improperly connected	Electrical check
102.	Insulating sleeves omitted from internal wiring connections	Visual
103.	Excessive solder used in soldering wires to receptacle	Visual
104.	Receptacle improperly oriented	Visual
105.	Marking misleading or unidentifiable	Visual
106.	Any flat head screw protruding	Visual
107.	Rubber gaskets torn, twisted, missing or otherwise defective	Visual
108.	Any of the protective caps missing or cracked	Visual
109.	Incorrect height of pins	Visual*
110.	Objective evidence relative to material missing or improper (see 3.1)	Visual
111.	Evidence of poor workmanship (see 3.15)	Visual
Minor:		
201.	Serialization improper or missing (see 3.4)	Visual

*In the event the pin(s) do not appear to be within 1/32 inch in height relative to the adjacent pins, gage (see 4.3.4.1) shall be used to determine the variation from the normal

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4.3.3.1 Continuity resistance, insulation resistance, pneumatic system sealing, operational tolerance, and repeatability. Each BSA shall be tested in accordance with 4.4.1 through 4.4.10. If any item fails to comply with 3.7 through 3.11, the item shall be rejected and removed from the lot.

4.3.3.2 Overpressure, underpressure, acceleration, mechanical shock, pressure cycling, transportation vibration, and operational vibration. A sample shall be selected at random in accordance with Table V. The sample shall be tested in accordance with 4.4.11. If any item fails to comply with 3.11.1 through 3.11.7, the lot shall be rejected.

Table V

Sample sizes for destructive testing

Number Required to be Delivered	Sample Size for No Failures
2-8	1
9-180	2
181-500	3
501-800	5
801-1300	7
1301-3200	10

Lot Size Required = Number Required to be Delivered + Sample Size.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (See EL-8871314). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on EL. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

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4.4.4 Pneumatic system sealing at -65°F. After the temperature of the BSA has stabilized at -65 ± 5°F. (see 6.4), follow the procedure outlined in 4.4.3.

4.4.5 Operating tolerance. Connect the BSE to the electrical circuit shown in Figure 1 and evacuate the pneumatic system to a pressure 15 mm Hg greater than the expected operating point. Continue to evacuate at a rate slow enough to be compatible with test equipment and/or personnel until all elements close. Record the operating points and return to ambient pressure. Repeat until three readings for each element are obtained. All operating points must be within the tolerance envelope shown in Table VI.

Table VI

Operating tolerance points

BSE Nominal Altitude Setting	Nominal Pressure (mm Hg)	Max. Allowable Operating Error (mm Hg)	Repeatability (mm Hg)
35,000 Ft.	178.8	± 10.45	2.0

4.4.6 Operating tolerance at -65°F. Place the BSA in a test chamber and lower the chamber temperature to -65 ± 5°F. After the chamber temperature has stabilized (see 6.4) and with temperature of the BSA still at -65°F. determine the operating tolerance of each BSE using the method of 4.4.5.

4.4.7 Operating tolerance at 160°F. The BSA shall be tested as specified in 4.4.6 except the environmental temperature shall be 160 ± 5°F. instead of -65 ± 5°F.

4.4.8 Repeatability. Determine the repeatability of operation for each BSE from the test results of 4.4.5.

4.4.9 Repeatability at -65°F. Determine the repeatability of operation for each BSE from the test results of 4.4.6.

4.4.10 Repeatability at 160°F. Determine the repeatability of operation for each BSE from the test results of 4.4.7.

4.4.11 Environmental testing (see 3.11).

4.4.11.1 Overpressure. Determine the average operating point pressure of each BSE in accordance with 4.4.5, except that 5 dives shall be employed. Subject the pneumatic system of the BSA to a pressure of 820 mm Hg for 20 hours. Immediately at the end of the overpressure period, redetermine the average operating point pressure of each BSE, using 5 dives. For each BSE determine the difference between the operating pressure averages for before and after the over pressure period.

4.4.11.2 Underpressure. Determine the average operating point pressure of each BSE in accordance with 4.4.5, except that 5 dives shall be employed. Subject the pneumatic system

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Table VII

Operation vibration tape characteristics

Time (sec)	Type of Signal	Ratio of Test Level To Calibration Level	RMS g Level
0-20	Noise (calibration (1)	1	Set gain to give 1.1 g RMS
20-30	None	-	-
30-31	Noise (1)	11	12
31-46	Noise (1) plus sinusoid (2)	4.25	4.6
	Noise alone	3.75	4.1
	Sinusoid alone	2.	2.2
46-61	Noise (1) plus sinusoid (3)	8.5	9.2
	Noise alone	7.5	8.2
	Sinusoid alone	4.	4.4
61-99	Noise (1) plus sinusoid (4)	4.25	4.6
	Noise alone	3.75	4.1
	Sinusoid alone	2.	2.2
99-100	Noise (1)	11.	12.

(1) Band limited (15-1500 Hz) white Gaussian noise.
(2) Sine wave swept from 800 to 1300 Hz at a rate increasing directly with frequency. One full sweep in 15 seconds.
(3) Sine wave swept from 1300 to 800 Hz at a rate decreasing directly with frequency. One full sweep in 15 seconds.
(4) Sine wave swept from 15 to 1500 Hz and back to 15 Hz at a rate increasing directly with frequency. One full cycle in 38 seconds.

For creating the random noise, the output of a General Radio random noise generator, or its equivalent, may be used if proper care is taken to insure the correct amplitude distribution of the signal. For testing the noise generator or any other associated equipment, the use of an amplitude distribution analyzer is suggested. For insuring the correct noise bandwidth, a filter with an asymptotic slope of at least 24 decibel (Db) per octave and 3 Db points at 15 and 1500 Hz shall be considered acceptable. Each BSE shall be monitored continuously during the vibration test for contact chatter using the procedure specified in 4.4.11.7.2. Following the vibrations the BSA shall be tested for 3.6, 3.7, 3.8 at ambient temperature, and 3.9, at ambient.

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chamber and returned to ambient temperature without forced warming. Excess moisture, if present, may be removed by blowing dry, filtered compressed air over the unit for a period of five minutes. The covers shall be removed and the BSA inspected for evidence of deterioration. In addition, the BSA shall be tested for 3.6, 3.7, 3.8 at ambient temperature, and 3.9, at ambient.

4.4.12 Drift and leakage. Perform the drift and leakage test using the procedure specified on dwg 8853654.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The BSA shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The BSA shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The BSA shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 3.5). Data specified in 3.4, 3.14 and 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801 ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

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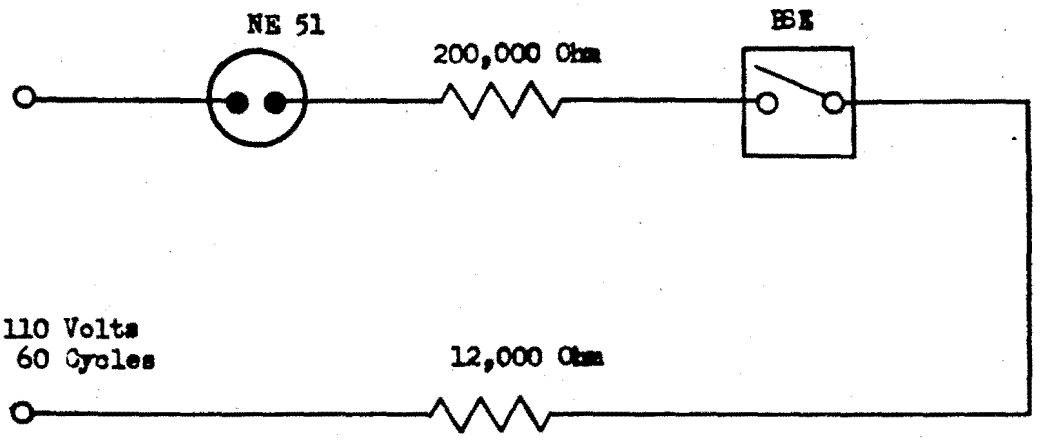


Figure 1. Acceptable Test Circuit To Determine Operating Point

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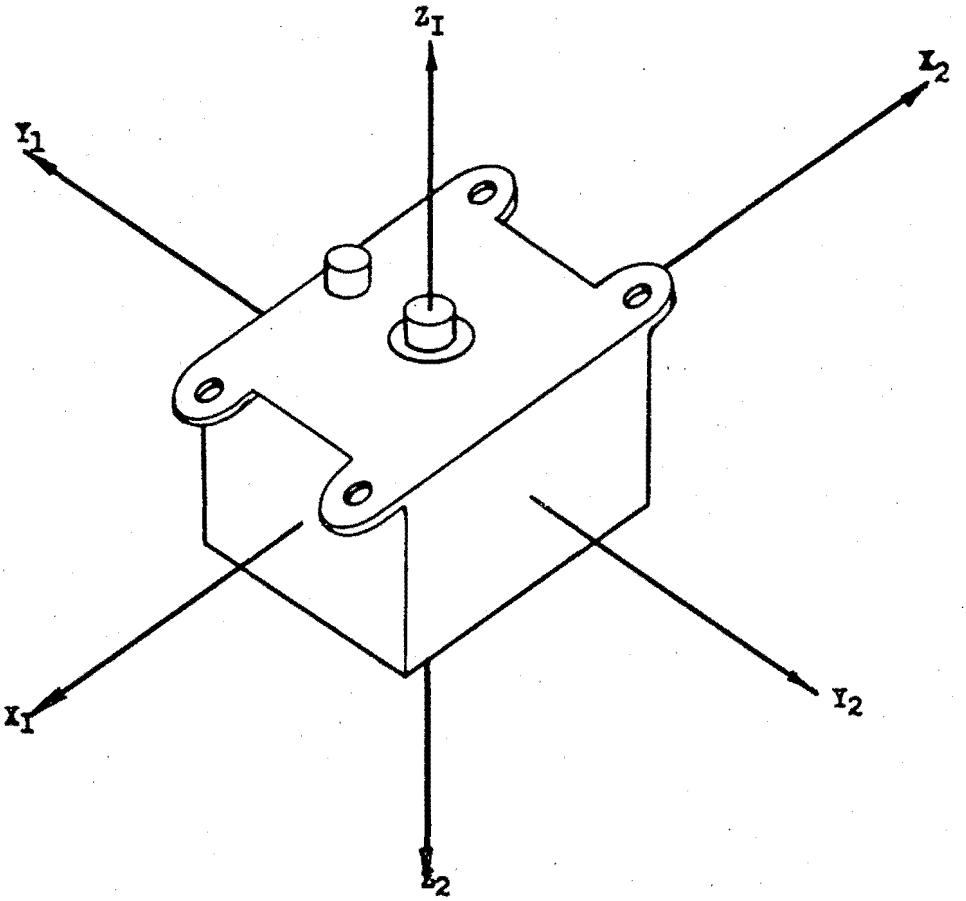


Figure 3. Axes To Be Used In Vibration And Acceleration Tests

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